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Project Title:	The Impact of Lunar Regolith on Nitrogen Fixation in a Plant Growth Promoting Rhizobacterium		
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Task Description:	Bioregenerative life support systems will be essential for advanced space exploration. A key component of bioregenerative life support systems is the ability to grow plants in extraterrestrial environments like the Moon using in situ resources such as lunar regolith. Recent studies have shown that Arabidopsis thaliana can grow in lunar regolith, illustrating the potential of lunar regolith as a growth substrate for plants and their associated rhizosphere. Nevertheless, one of the biggest hurdles to growing plants on the Moon will be the ability to provide plants with bioavailable nitrogen. Most of the nitrogen fertilizer used on Earth comes from the Haber Bosch process, but the large amount of natural gas and infrastructure needed to carry out this process make it difficult to reproduce in extraterrestrial environments. Plant growth promoting rhizobacteria (PGPR) also provide plants with bioavailable nitrogen through a process known as biological nitrogen fixation. However, to harness this ability, we need to understand how lunar regolith impacts PGPR and their ability to carry out nitrogen fixation. Our goal is to determine how lunar regolith impacts the physiology of Rhodopseudomonas palustris, one of the most prolific nitrogen-fixing PGPR known.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will provide insights into the potential of bacteria to access essential metals for nitrogen fixation from extraterrestrial environments and contribute to the development of sustainable life-support systems for long-duration space missions.
Task Progress:	Scientific goals/objectives of the proposed research Bioregenerative life support systems will be essential for advanced space exploration. A key component of bioregenerative life support systems is the ability to grow plants in extraterrestrial environments like the Moon using in situ resources such as lunar regolith. Recent studies have shown that Arabidopsis thaliana can grow in lunar regolith, opening up the exciting possibilities for plant growth on the Moon. However, one of the biggest hurdles to growing plants on the Moon will be the ability to provide plants with bioavailable nitrogen since lunar regolith has very little nitrogen. Most of the nitrogen fertilizer used on Earth comes from the Haber-Bosch process, but the large amount of natural gas and infrastructure needed to carry out this process make it difficult to reproduce in extraterrestrial environments. Plant growth promoting rhizobacteria (PGPR) offer a solution to this problem since they can fix nitrogen biologically. However, to harness this ability, we need to understand how lunar regolith impacts biological nitrogen fixation. The primary goal of this proposal is to conduct a comprehensive investigation into the impact of lunar regolith on the ability of a PGPR to fix nitrogen. This research will provide insight into potential limitations to nitrogen fixation in lunar regolith and may uncover crucial insights to enable the development of sustainable life support systems for future space exploration missions. The key enzyme involved in biological nitrogen fixation is nitrogenase. The most efficient and widely distributed nitrogenase requires both molybdenum and iron for its unique MoFe cofactor in its active site. Analysis of lunar regolith taken from the Apollo 11 and Apollo 12 missions indicate that there is almost 30 times less Mo in lunar regolith than in terrestrial soil. This means that metal availability, including trace metals, may impact our ability to harness nitrogen fixation in lunar regolith. Additionally, metals such as aluminum are much more abundant in regolith from the Highland region on the Moon, and increased aluminum oxide in soil has been shown to correlate with a decrease in the distribution of nitrogen-fixing bacteria and reduce nitrogen fixation in root nodules. The objective of this proposal is to determine how metal availability in lunar regolith impacts nitrogen fixation by a PGPR. This will be done through the following objectives: Aim 1. Determine if lunar regolith simulant impacts viability of the PGPR, Rhodopseudomonas palustris, under nitrogen-fixing conditions. Aim 2. Determine how metal availability in lunar regolith simulant impacts activity of nitrogenase. Research accomplishments For this study, we utilized three distinct lunar regolith simulants: OPRH4W30, a representation of the lunar Highlands regolith; OPRLJSCN, with a composition comparable to JSC-1A; and OPRL2N, which exhibits geotechnical properties, including cohesion and particle size distribution, closely approximating those of Mare lunar regolith. To evaluate the effect of the lunar regolith simulants on the growth and viability of R. palustris, we cultured the bacterium in media supplemented with increasing concentrations of simulants (0, 1, 10, 100, and 150 g/L). The cultures were incubated for eight days, and biomass production was measured under both nitrogen-fixing and non-nitrogen-fixing conditions. Under nitrogen-fixing conditions, the presence of the simulants did not result in a significant impairment of bacterial growth, and nitrogenase activity remained comparable across conditions. In contrast, under non-nitrogen-fixing conditions, a slight reduction in growth was observed at higher concentrations of OPRH4W30, while the other simulants did not elicit a significant inhibitory effect. To gain insight into how lunar regolith simulant influences R. palustris physiology, RNA sequencing (RNA-seq) was performed to assess gene expression changes in cultures grown with and without the highest concentration of simulant. The strongest transcriptional response was observed in the presence of OPRH4W30, whereas OPRLJSCN and OPRL2N induced fewer differentially expressed genes. Notably, genes involved in iron homeostasis were differentially expressed across all simulant conditions, including those associated with siderophore production, iron uptake, and iron-sparing mechanisms. The widespread downregulation of these genes suggests that R. palustris experiences iron-rich conditions in the presence of simulants. This conclusion is further supported by a comparison of gene expression patterns in R. palustris grown without simulant but supplemented with excess iron, indicating that iron availability is sufficient under all tested conditions. Regarding nitrogen fixation, genes encoding an alternative nitrogenase that incorporates vanadium instead of molybdenum were upregulated across all simulant conditions. This expression pattern is consistent with the presence of vanadium detected in all simulants and aligns with observations that vanadium supplementation induces similar gene upregulation in the absence of regolith. These findings suggest that R. palustris detects and responds to vanadium in the simulants. The OPRH4W30 condition elicited the greatest transcriptional changes. In this condition, we observed increased expression of genes related to oxidative stress, DNA repair, and sulfur assimilation. Additionally, genes involved in molybdenum uptake were upregulated, suggesting R. palustris experiences molybdenum limitation in OPRH4W30. This was unexpected, as no significant differences in molybdenum availability were detected between simulants. However, OPRH4W30 contains twice the aluminum concentration of the other simulants. Aluminum exposure has been shown to

	impair nitrogen fixation in root-nodulating bacteria and induce oxidative stress in other bacterial species. These findings suggest that R. palustris experiences increased stress in OPRH4W30, likely due to elevated aluminum levels. While nitrogenase activity and overall growth were not significantly impacted under these conditions, additional environmental stressors—such as reduced atmospheric pressure, microgravity, or radiation—could further compromise nitrogen fixation by plant growth-promoting rhizobacteria (PGPR) in Highland regolith compared to Mare regolith. Nitrogen is a critical element for life and essential for plant growth, which is necessary for food production. Nitrogen fixation on the Moon could potentially provide a sustainable source of nitrogen for future lunar settlements or long-duration missions. However, it is unclear if the metals required for nitrogen fixation are bioaccessible in lunar regolith. This study is the first to provide insights into the potential of nitrogen-fixing bacteria to access essential metals for nitrogen fixation from lunar regolith simulants. We have also shown that expression signatures can provide insight into how the composition of lunar regolith signature could impact physiology of a PGPR.
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